COLLEGE OF COMPUTING AND INFORMATICS

TEAM 1 · THE 60-SECOND PITCH

# RAEJ: Translating Languages No LLM Knows

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## THE PROBLEM

Thousands of low - resource languages and conlangs have a dictionary and a grammar but no parallel text, so standard machine translation can't serve them

## TARGET IMPACT

Accuracy on a 25 -sentence validated gold set, and prompt tokens per request: RAG vs. the full ~28k -token context.

## CORE TECH STACK

Python · FastAPI · PostgreSQL + pgvector · LangChain · spaCy · Nomic Embed · LM Studio / Ollama / OpenAI API · Docker